# Temperature/Process Digital Indicators

1 YEAR WARRANTY

YEAR 2000 COMPLIANT

**Model DP21-TC**

**$299**

**Shown**




**Shown larger than actual size**

- **±0.25% High Accuracy Indication**
- **User-Selectable Inputs (Thermocouples)**
- **User-Selectable Ranges**
- **Programmable Scaling (mV dc, mA dc)**
- **Power Supply: 100 to 240 Vac or 24 Vac/dc**
- **Versatile Alarm Configurations**
- **Standard Min/Max Recall**
- **Transducer Power Supply Available**
- **Optional RS-232C, RS-422A, or RS-485 Interface**
- **Analog Output Scaling Available**

The DP20 series indicators, available for thermocouple, RTD, voltage or current inputs, feature a large, easy-to-read, 4-digit LED display. This indicator has programmable multi-range inputs and a high accuracy of ±0.25% plus one digit. The unit also offers fast response of 0.25 seconds, decimal point setting, plus scaling lower and higher limit setting. It features minimum and maximum reading storage, for later recall. Available options are normal or latching relay alarms, 4 to 20 mA, 0 to 10 Vdc and 0 to 10 mV analog output. The DP20 operates on universal power supply or 24 Vac/Vdc. Other options include a communications interface and transducer power supply.

## Specifications

**Display**

**Range Display:**
7-segment LED (4-digit)

**Display Accuracy:**
±0.25% + 1 digit of measuring range

**Display Cycle Range:**
0.25 sec or 2 sec
(by internal switch)

**Legend Change:**
°C and °F (by internal switch)

**Monitor Display:**
Max., min, hold, communication, alarm and range

**Setting**

**Setting Method:** By front key

**Setting Keys:** Display, communication, alarm, range, setting, shift and up

**Setting Type:**
Position of decimal point, display scaling, sensor compensation, analog output scaling, alarm, alarm sensitivity, latching OFF, communications (interface), etc.

**Input**

**Sampling Cycle:** 0.25 second

**Thermocouples:** J, K, T, E, R, S , B, C, J DIN and T DIN

**Input Impedance:** 500 kΩ min.

**Lead Wire Resistance:**
100 Ω max.

**Sensor Compensation:**
±99.9 or ±9.99 unit
(depending on input range)

**Burnout Scaling:**
Up-scale (standard feature)

**RTD:** 3-wire, Pt, 100 Ω,
both 0.00385 and 0.00392 curves

**Excitation Current:** 0.6 mA

**Lead Wire Resistance:**
5 Ω max. per wire

**Sensor Compensation:**
±99.9 or ±9.99 unit
(depending on input range)

**Voltage:** 0 to 10 mV, 0 to 50 mV, or 0 to 100 mV dc changeable; 0 to 1 V, 0 to 5 V, 0 to 10 Vdc changeable

**Input Impedance:** 500 kΩ min

**Current:**
4 to 20 mA or 0 to 20 mA dc changeable

**Input Impedance:** 250 Ω

**Display Scaling:** Only voltage and current inputs available (-1999 to +9999 minimum span 100 counts, maximum span 10,000 counts)

**Isolation:** Between input and internal circuit

## Optional Functions

**Alarms**
**Operating Mode:** Individually set for high/low and deviation or absolute action
**Setting Range:** -1999 to 9999
**Sensitivity:** 0.2 to 9.9 or 2 to 99 by front key
**Mechanical Relay:**
SPDT, 250 Vac, 2.5 A/Resistive load
**Inhibit/Non-Inhibit:** Selectable by internal switch

**Analog Output**
**Output:** 0 to 10 mV dc (output resistance 10 Ω)
0 to 10 Vdc (load current 2 mA max.)
4 to 20 mA dc (load resistance 300 Ω max.)
**Accuracy:** ±0.25% FS of display value
**Resolution:** Approx. 0.0125 FS (1/8000)
**Isolation:** Isolated between input and internal circuit

**Interface**

**Communications**
**Type of Interface:** RS-232C, RS-422A or RS-485
**Speed:** 1200, 2400, 4800 or 9600 bps selectable
**Data Bit:**
Selectable 7-bit odd-parity or 8-bit without parity
**Address:** 0–31
**Code:** ASCII code
**Isolation:** Isolated between input and internal circuit

**General**
**Transducer Power Supply:** 24 Vdc (±3 V), 50 mA
**Power Supply:** 100 to 240 Vac, 50/60 Hz or 24 Vac/dc
**Power Consumption:** 8 VA, 24 Vdc 8 W
**Insulation Resistance:** 500 Vdc 20 MΩ between input terminal and earth terminal. 500 Vdc 20 MΩ between power supply terminal and ground terminal
**Dielectric Strength:** One minute at 500 Vac between input terminal and earth terminal. One minute at 1500 Vac between power supply terminal and earth terminal
**External Dimensions:** 48 H x 96 W x 110 mm D (1.89 x 3.78 x 4.33")
**Panel Cutout:** ⅛ DIN, 45 x 92 mm (1.772 x 3.622")
**Weight:** 300 g (0.66 lb)




**KMTSS-125G-6 Thermocouple, $24.**
**Probe is sold separately. See Section A.**



## Input Code and Range Table

| Input Code | Input Type | Range °C | Range °F |
|---|---|---|---|
| TC | J | -100 to 600.0 | -148 to 1112 |
| | K (high) | -200 to 1200 | -328 to 2192 |
| | K (low) | -100.0 to 800.0 | -148 to 1472 |
| | T | -199.9 to 200.0 | -328 to 392 |
| | E | 0.0 to 700.0 | 32 to 1292 |
| | R | 0 to 1700 | 32 to 3092 |
| | S | 0 to 1700 | 32 to 3092 |
| | B | 400 to 1800 | 750 to 3300 |
| | C | 0 to 2300 | 32 to 4172 |
| | J DIN | 0.0 to 600.0 | 32 to 1112 |
| | T DIN | -199.9 to 200.0 | -328 to 392 |
| RTD | 100 Pt,0.00385 | -199.9 to 600.0 | -328 to 1112 |
| | and 0.00392 | 0.00 to 99.99 | 32.0 to 212.0 |
| MV | Low Level Voltage | 0 to 10 mV*<br>0 to 50 mV*<br>0 to 100 mV* | |
| V | High Voltage | 0 to 1 V*<br>0 to 5 V*<br>0 to 10 V* | |
| MA | Current | 0 to 20 mA*<br>4 to 20 mA* | |

** User scalable; range is -1999 to 9999 counts, min. span is 100 counts, max. span is 10,000 counts*

## To Order *(Specify Model Number)*

| Model No. | Price | Description |
|---|---|---|
| DP21-(*) | $299 | Digital panel indicator |

***Specify Input Type:** **TC** – Thermocouple, **RTD** – Pt RTD 3-wire 100 Ω, **MA** – Current to 20 mA dc, **MV** – Voltage to 100 mV, **V** – Voltage to 10 V*
***Ordering Example: DP21-TC-AL**, meter with thermocouple input and optional dual alarms, $299 + 100 = **$399***

## Options

| Ordering Suffix | Add'l Cost | Description |
|---|---|---|
| -AL | $100 | Dual alarms |
| -ALL | 120 | Dual latching alarms |
| -AO1[1] | 60 | 0 to 10 mV dc recorder output |
| -AO2[1] | 60 | 4 to 20 mA dc recorder output |
| -AO3[1] | 60 | 0 to 10 Vdc recorder output |
| -RS232[1] | 120 | RS-232 communications |
| -RS422[1] | 120 | RS-422 communications |
| -RS485[1] | 120 | RS-485 communications |
| -XP | 50 | Transducer power supply |
| -24VDC[2] | 50 | 24 Vdc power supply |
| -24VAC | 50 | 24 Vac power supply |

[1]*Only one of these options can be installed per unit.*
[2]*When computer interface options are ordered, "**-24VDC**" (dc power supply) is not available.*





For Additional Controllers, See Section P

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